BIOL-1 Practice Test 02 — Answer Key

Modules 5-6: Membranes and Metabolism

Part A: Multiple Choice Answers

Q	Answer	Explanation
1	B	The fluid mosaic model describes membrane structure
2	C	Cell membrane = phospholipid bilayer
3	B	Phospholipids have hydrophilic heads and hydrophobic tails
4	B	Cholesterol helps maintain membrane fluidity
5	D	Active transport requires ATP
6	B	Hypotonic = water moves INTO cell
7	C	Isotonic = no net water movement
8	B	Enzymes lower activation energy
9	B	ATP is the energy currency of cells
10	B	Substrates bind at the active site
11	B	Denaturation = loss of enzyme shape
12	B	ATP releases energy via phosphate hydrolysis
13	B	Enzymes can be reused (not consumed)

Part B: Fill in the Blank Answers

Q	Answer
14	fluid
15	osmosis
16	active
17	hypotonic
18	substrate
19	kinetic
20	competitive
21	metabolism
22	catabolic

Part C: Short Answer Key

23. Passive vs. Active Transport:

- Passive transport: does not require energy (ATP); moves with concentration gradient (ex: diffusion, osmosis, facilitated diffusion)
- Active transport: requires energy (ATP); moves against concentration gradient (ex: sodium-potassium pump)

24. Cell in Hypertonic Solution:

- Water moves OUT of the cell (toward higher solute concentration)
- Animal cell: will shrink (crenation)
- Plant cell: will plasmolyze (cell membrane pulls away from cell wall), but cell wall prevents complete collapse

25. Enzyme Function & Specificity:

- Enzymes speed up reactions by lowering the activation energy required for the reaction to proceed.
- They are specific because the shape of their active site matches only specific substrate molecules (lock and key / induced fit).

26. ATP Structure & Function:

- Structure: Adenine (nitrogenous base), Ribose (sugar), and 3 Phosphate groups.
- Function: Energy is stored in the bonds between phosphate groups; when the terminal phosphate bond is broken (hydrolysis), energy is released for cellular work.

End of Answer Key